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Systematically visualizing ChatGPT used in higher education: Publication trend, disciplinary domains, research themes, adoption and acceptance

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ABSTRACT

Since it was released in November 2022, ChatGPT has been exerting revolutionary influence on the realm of higher education. In order to obtain a comprehensive understanding of the research landscape, we conduct a systematic literature review on the studies of ChatGPT used in higher education. Both quantitative and qualitative methods were adopted to bibliometrically examine the included literature selected from Web of Science and Scopus through the PRISMA protocol. Tools of VOSviewer and CitNetExplorer were employed to visualize the citation information. Our findings showed that the recent two years witnessed an ever-growing popularity of this research theme. Citation information analysis reveals the most influential authors, countries, sources, organizations and four focused topics. The disciplinary distribution of related research indicates a wide range of categories. More importantly, ChatGPT was found to be versatile in assisting teachers, students and researchers with a variety of tasks, and the factors influencing the acceptance of this technology among college students could be investigated through models like TAM, UTAUT and their extensions. We suggest future studies to focus on the ways to address the limitations and ethical issues of ChatGPT through AI literacy cultivation and joint efforts of all stakeholders.

1. Introduction

As one of the most representative technologies in Industry 4.0, generative artificial intelligence (Gen AI) has taken the world by storm in recent years. ChatGPT, an generative pre-trained transformer that can produce human-like text and conduct interactive conversation, is one of the most representative Gen AI products. Since its advent in November 2022, it has been gaining popularity and exerting its influence on the realm of higher education. Compared to previous educational technologies, ChatGPT stands out for its interactive and natural language processing (NLP) capabilities. Its vast pre-trained data on extensive publicly available sources and reinforcement learning from human feedback made it a useful tool to perform various language processing tasks (Alqahtani et al., 2023). This feature significantly enhances user engagement and accessibility, enabling more intuitive and conversational learning experiences. Besides, ChatGPT can potentially provide personalized learning experiences tailored to individual students' needs and abilities (Azaiz, Deckarm, & Strickroth, 2023; H. Lee, 2023). This represents a significant advancement over traditional educational tools, which generally offered non-specific content.

The significance of ChatGPT in higher education is highlighted in its capabilities to assist teachers, students, and researchers with various educational tasks (Ansari, Ahmad, & Bhutta, 2023). Specific case studies have demonstrated the revolutionary impact of ChatGPT on higher education since its release. On the one hand, ChatGPT significantly boosts educational support and accessibility. Its widespread availability offers comprehensive educational assistance, thereby reducing educational barriers and fostering a more inclusive learning atmosphere. For example, A university in the United Arab Emirates has integrated ChatGPT into its library service to provide 24/7 access to academic support. Students and faculty who need academic support and service outside the regular working hours can interact with the ChatGPT-powered robot to receive instant feedback and explanations (Lappalainen & Narayanan, 2023). On the other hand, ChatGPT encourages creativity and critical thinking, engaging learners with innovative approaches and analytical mindsets. A chemistry course has incorporated ChatGPT into its curriculum to encourage students to engage in AI-based activities. Significant improvements were noticed in students' confidence to ask insightful questions, analyze information, and comprehend complex concepts (Guo & Lee, 2023). These case

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studies demonstrate how ChatGPT has revolutionized higher education by enhancing learning support, driving innovation in teaching methods, streamlining administrative tasks, and fostering critical thinking and creativity among college teachers and students.

Meanwhile, concerns and controversies regarding the challenges and potential risks came along. The academia worried that the use of ChatGPT would lead to more plagiarism in scholarly writing (Singh, 2023). And the hallucination issue, which refers to generating non-existent or incorrect content, raised people's concerns on its liability (Alqahtani et al., 2023). Challenges also lie in privacy infringement, algorithmic bias, ethical considerations and so forth (Bannister, Urbieta, & Penalver, 2023; Dempere, Modugu, Hesham, & Ramasamy, 2023; Farrelly & Baker, 2023).

Current research on ChatGPT in higher education, while making progress, still exhibits several notable gaps. Since potential risks and ethical considerations come along with the emergence of ChatGPT, some research overemphasized the negative impact while neglecting ChatGPT's diverse applications and achievements in educational sector. Secondly, in view of the ever-growing popularity of Gen AI, the existing literature reviews failed to catch up with the latest studies, necessitating inclusion of new publications. Thirdly, there is a lack of evidence to explain the causes behind ChatGPT's varying performance across different subjects. Finally, while existing studies have explored ChatGPT's usage in higher education from different perspectives, they have not yet done an in-depth investigation of each single theme.

In order to address these research gaps, this systematic literature review will focus on the latest high-quality academic publications pertaining to ChatGPT used in the context of higher education. Bibliometric approaches will be employed to analyze the publication trend, the most focused disciplinary domains, the main themes and the acceptance models. By doing so, it aims to provide a comprehensive understanding of the current research landscape. This will not only facilitate the identification of the most critical issues but also suggest directions for future studies.

2. Literature review

Table 1 shows four existing literature reviews investigating ChatGPT used within the context of higher education ranked in chronological order. Following the guideline proposed by Kraus et al. (2022), we categorized the included four reviews into systematic literature review (SLR) and non-systematic literature review (non-SLR) considering whether there was a protocol for guiding data curation and analysis. In these reviews, ChatGPT is sometimes synonymously referred to as "generative artificial intelligence", "Gen AI" or "large language model (LLM)". Given the fact that the four reviews were written in earlier 2023, it is essential to incorporate more recent advancements to provide a more comprehensive view of current academic landscape. To achieve this, we aim to include the most recent high-quality publications and analyze their features, so as to address the following research questions.

Table 1
Previous reviews on ChatGPT-assisted higher education.

Author(s), year	Number and time span of the reviewed publications	Types and methods	Theme
Farrelly and Baker (2023)	around 50; from the release of ChatGPT to September 2023	non-SLR, critical analysis	potential of Gen-AI on higher education and its impact on academic integrity
Bannister et al. (2023)	68; from the release of ChatGPT to April 2023	SLR, thematic analysis	implications of Gen-AI in higher education
Dempere et al. (2023)	92; from the release of ChatGPT to April 16, 2023	SLR, thematic analysis	explore the impact of ChatGPT in higher education by descriptive and thematic analysis
Ansari et al. (2023)	69; from the release of ChatGPT to July 12, 2023	SLR, bibilometric analysis	the use of ChatGPT in higher education and its motivations and concerns

RQ1: What is the publication trend of the literature in the field of ChatGPT-assisted higher education since the launch of ChatGPT?
RQ2: Who are the top 10 authors, sources, organizations and countries based on citation data in the field of ChatGPT-assisted higher education?
RQ3: What disciplinary domains have current studies mainly engaged with?
RQ4: What are the major themes discussed by current publications?

As a non-systematic review, Farrelly and Baker (2023) critically explored the multifaceted impact of Gen AI on higher education. They believe that ChatGPT, as a rising star in AI family, is more disruptive to higher education and society than previous inventions. Since the overall research was not systematically conducted, their research findings are far from convincing.

Adopting the thematic analysis approach, Bannister et al. (2023) touched on a wide rage of themes regarding AI's applications and implications in higher education, including language instruction, student learning, assessment and feedback, research activity, academic integrity and policy. Dempere et al. (2023) generally discussed both the advantages and potential risks of integrating ChatGPT into higher education. Ansari et al. (2023) mapped the global evidence of ChatGPT's applications in higher education, offering an insight into the contextual, methodological and disciplinary scopes of this research field. Although these reviews provided a initial overview of Gen AI's current status in higher education, a more in-depth investigation needs to be carried out to elucidate each research theme.

In summary, existing reviews, while shedding light on the overall trend and disciplinary distribution of ChatGPT-related research in higher education, left some important issues under-explored, such as the major themes discussed, the specific roles of ChatGPT, and the dynamics of user acceptance. To fill these gaps, we propose the following research questions.

RQ5: How can ChatGPT be effectively integrated into the learning, teaching and research processes in higher education?
RQ6: How does the performance of ChatGPT vary across different disciplines?
RQ7: What is the extent of adoption and acceptance of ChatGPT among college students?

These questions will focuses on mining out the existing research themes, summarizing the specific roles of ChatGPT in assisting college teaching, learning and research and exploring the factors influencing user's adoption and acceptance of ChatGPT in higher education, areas that previously received limited attention.

3. Methods

3.1. Research design

As shown in Fig. 1, we aimed at carrying out the systematic review with the guidance of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) 2020 (Page et al., 2021), a protocol widely recognized for providing a robust framework for conducting systematic reviews. VOSviewer and CitNetExplorer, two powerful science mapping tools, were employed to visualize and analyze bibliometric networks. VOSviewer is a software developed for constructing bibliometric maps based on the metadata from the included publications. Its versatile display techniques enable researchers to view a map in full detail (van Eck & Waltman, 2010). Similarly, CitNetExplorer

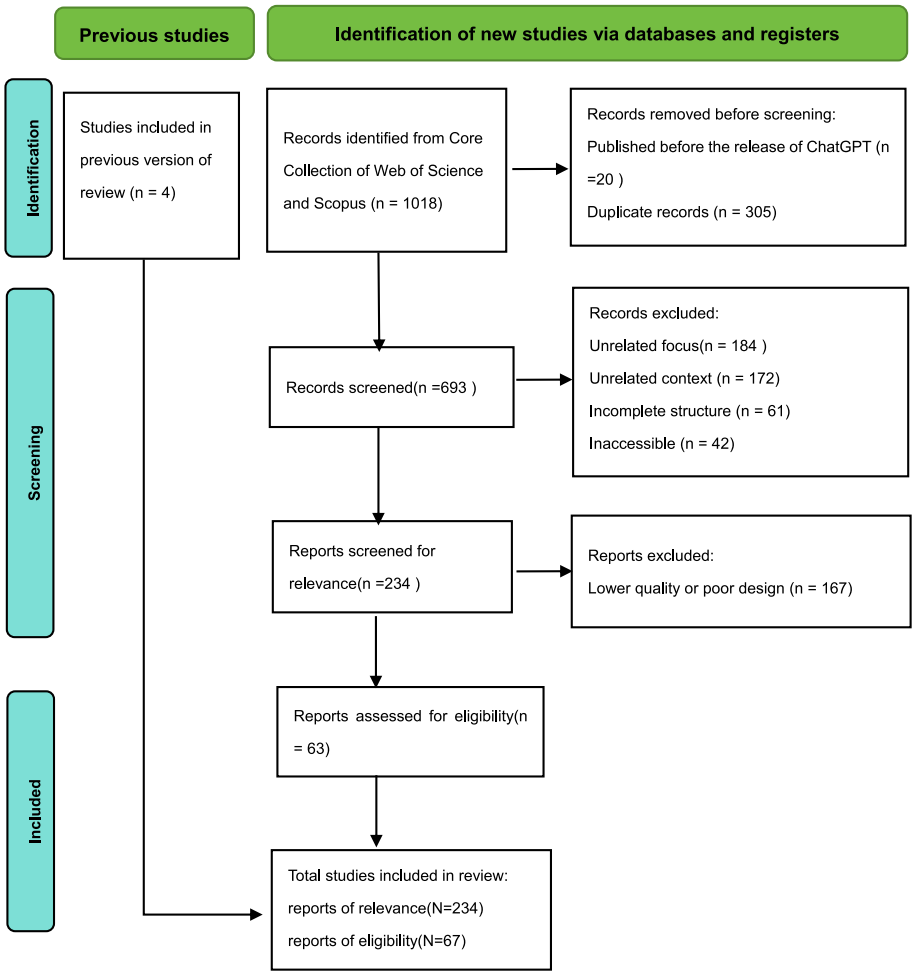


Fig. 1. A PRISMA flowchart of literature search and screening.

offers detailed visualizations of citation networks, allowing users to identify key researchers and research areas and analyze the structure and dynamics of scientific fields, capturing the development of a research field over time (van Eck & Waltman, 2014). VOSviewer and CitNetExplorer significantly enhanced the analysis of citation information by generating comprehensible and analyzable citation networks, offering deeper insights into the scientific landscape of a given research area.

3.2. Databases and search strategy

We conducted our literature search on March 14, 2024, utilizing Web of Science (the Core Collection) and SCOPUS, two of the most authoritative and comprehensive literature databases, as our searching resources. We attempted to identify relevant articles by imputing search strings with Boolean operators. The first one is "ChatGPT OR GPT* OR "Open AI" OR "Generative AI" OR "Generative Artificial Intelligence" OR "Gen AI" OR "large language model*"" with the Titles; the second one is "college OR university OR "higher education" OR "higher institut*" OR "Tertiary Education"" with the Topics. They were jointly used to ensure that all peer-reviewed texts with regard to ChatGPT used in higher education were possibly collected. As a result, a total of 1018 peer-reviewed articles were identified.

3.3. Inclusion and exclusion criteria

Table 2 shows the inclusion and exclusion criteria to filter out papers relevant and eligible for our research in a three-step approach. In the first step was to exclude articles published before November 30th, 2022, the date of ChatGPT-3.5 emergence and duplicate publications. In the second step, we restricted the criteria to obtain the most relevant articles which (1) focus on the use of ChatGPT or other large language models, (2) pertain to the background of higher education, (3) have complete components of an academic article, including abstract, introduction, methodology, findings, discussion, references and so on, and (4) have accessible Digital Object Identifier (DOI). Consequently, 234 relevant articles were filtered, which would be specifically used to answer RQ1 to RQ4.

Table 2
Inclusion and exclusion criteria.

Step	Criterion	Inclusion	Exclusion
1	Time period	November 30th, 2022 to March 14th, 2024	before the emergence of ChatGPT
2	Topic	focusing on the use of ChatGPT or other LLMs	unrelated focuses
	Education Sector	related to higher education	unrelated to higher education
	research structure	with complete components of an academic article	Incomplete structure inaccessible
	accessibility	accessible DOI	inaccessible
3	Standards for Reporting on Empirical Social Science Research(2006)	with a high average score	with relatively lower average score

Subsequently, we adopted Standards for Reporting on Empirical Social Science Research in AERA Publications: American Educational Research Association (Association, 2006) to assess the quality of the relevant publications. In the evaluation process, scores were given to each relevant article from eight aspects, namely problem formulation, design and logic, source of evidence, measurement and classification, analysis and interpretation, generalization, ethics, title, abstract and headings. Two trained and experienced reviewers participated in the quality assessment with rigorous adherence to the established criteria, and a third rater was invited to help the decision making whenever there was a disagreement. The final inter-rater reliability reached a satisfactory level ($k = 0.886$). Furthermore, the four literature reviews previous analyzed were also included. Ultimately, 67 highly-rated articles (marked with asterisks in the references) were selected for further evidence extraction and analysis to answer RQ5 to RQ7.

3.4. Content analysis

In order to better address the research questions and achieve our research objectives, we conducted a content analysis of the selected studies. We extracted and analyzed the information both descriptively and thematically. The first focus is the publication trend of related publications. With the help of VOSviewer, we delved into the top authors, countries, sources, organizations of all 234 relevant publications. The specific disciplines in each document were identified and categorized and major themes among all relevant articles were synthesized through keywords clusters. As for the thematic analysis, the researchers read the 67 highly-rated articles to extract evidence pertaining to the applications and effectiveness of ChatGPT in real educational practice and its performance across different disciplines. Citation network analysis through CiteNetExplorer was conducted to visualize the citation relationship among studies dedicated to investigating the acceptance and adoption of ChatGPT in higher education.

4. Results

4.1. Publication trend (RQ1)

Since the research was conducted just 18 months after the emergence of ChatGPT, it is more appropriate to examine the publication trend of ChatGPT-assisted higher education on a monthly basis rather than an annual basis. Researchers meticulously identified the publishing time or early access date of each relevant document from the given information within the databases. In the case of unstated publishing time, the indexed time was utilized.

It was discovered that general publication trend went through a constant upward tendency with occasional minor fluctuations from January 2023 to March 2024, which reflected the ever-growing popularity of ChatGPT in the realm of higher education. The first article of this academic interest was published in February 2023, after which the number rose gradually with an increase of 2–3 in the subsequent five months, soaring to its first peak in July. After that, the upward trend continued, with the number almost doubling in October, indicating a sustained growth in the academic attention on ChatGPT in higher education.

This uprising tendency might be attributed to several factors: (1) the accumulation of empirical evidence on using ChatGPT in various educational settings, (2) the emergence of more critical reflections on its implications for education, and (3) the comparison with other newly-launched AI chatbots. Thus, it can be predicted that in the near future this academic momentum will remain strong due to more diversified applications of ChatGPT and deeper investigation into its dynamics within higher educational sector.

4.2. Citation information analysis (RQ2)

We use the citation analysis function of VOSviewer to obtain top 10 authors, countries, sources, organizations respectively (Table 3). For a given item, the citation links attribute indicates the number of links of an item with other items, while total link strength attribute indicates the total strength of the links of an item with other items (van Eck & Waltman, 2010). Therefore these values are significantly valuable in reflecting the academic influence of each item.

Among the top 10 most cited authors, it is not surprising that OpenAI, as the inventor company of ChatGPT, was the most frequently cited author among all the studies, followed by Rudolph, J., Venkatesh, V. and Dwivedi, Y. The scope of most cited countries ranged from North America, Europe, Asia African to Oceania, which reflected the widespread attraction for Gen AI technology. The top three countries with the biggest amount of documents were USA, China and Australia. Nevertheless, if ranked by citation value, this sequence becomes England, Australia and USA. The top 10 most cited sources and organizations were obtain through similar methods.

These citation analysis data offered us an insight into the academic influence of different authors, countries, sources and organizations, which are beneficial for researchers in searching and selecting literature in this regard.

4.3. Disciplinary distribution of ChatGPT-related researches in higher education (RQ3)

In order to summarize the disciplinary distribution of the filtered publications, we adopted International Standard Classification of Education (ISCED) framework (UNESCO, 2012) to classify the 234 relevant literature into different disciplinary domains where 7 broad groups and 17 education fields were identified, as displayed in Table 4.

Some documents were categorized into two or more groups due to their multidisciplinary nature. General education was the most heated area, followed by Health and welfare and Science. In these intensively studied areas, the scope of categories tended to be tremendously extensive. For example, the fields of medicine in Health and welfare encompassed a wide range of sub-disciplinary branches, including Dermatology, Obstetrics Gynecology, Rheumatology, Urology Nephrology, Oncology, Ophthalmology, Otorhinolaryngology, Surgery, Radiology, Nuclear Medicine, Medical Imaging, and General Internal.

However, studies were sparsely distributed in humanities and arts, services, engineering manufacturing and construction. In particular, they were barely found in fields like fine arts, law, agriculture, veterinary, transport services and security services.

4.4. Major topics discussed by the relevant literature(RQ4)

Fig. 2 demonstrates the co-occurrence network of all keywords encompassing author keywords and those derived from titles and abstracts. The network distinctly shows seven colored clusters pivoting around three core knots: “ChatGPT” “artificial intelligence” and “higher education”. By examining the keywords within one or two adjacent clusters, we identified four major themes (Table 5).

The first theme “applications of ChatGPT in teaching, learning and research in higher education”, aligning with the RQ5, highlights ChatGPT’s powerful text-generation and conversational capabilities. It explores how these capabilities can assist learners, educators and researchers with a variety of tasks, ranging from course design, making assessment to cultivating skills, fostering innovation and promoting communication.

The second theme, resonating with RQ6, involves empirical studies and data that evaluate and compare ChatGPT’s actual performance across different subjects.

The third theme focuses on researchers’ efforts in examining students’ adoption and acceptance of ChatGPT. This is achieved through

Table 3
top 10 authors, countries, sources, organizations based on co-citation data.

Rank	Top 10 authors	Citation	Total link strength	Rank	Top 10 countries	Document	Citation
1	OpenAI	98	409	1	England	18	212
2	Rudolph, J.	58	373	2	Australia	23	181
3	Venkatesh, V.	47	459	3	USA	49	160
4	Dwivedi, Y.	46	395	4	Saudi Arabia	9	41
5	Cotton, D.	43	306	5	China	24	36
6	Sallam, M.	42	224	6	India	8	31
7	Kasneici, E.	40	302	7	United Arab Emirates	8	25
8	Hair, J.	38	333	8	Malaysia	9	15
9	Tlili, A.	34	266	9	South Africa	8	12
10	Stokel-walker, C.	33	151	10	Germany	10	7

Rank	Top 10 sources	Document	Citation	Rank	Top 10 organizations	Document	Citation
1	Interactive Learning Environments	4	44	1	University of Tasmania	3	64
2	Australasian Journal of Educational Technology	7	24	2	Monash University	4	55
3	International Journal of Educational Technology in higher education	4	23	3	University of Economics in Katowice	3	41
4	Education Sciences	6	21	4	Charles Sturt University	5	32
5	International Journal of Human-Computer Interaction	5	18	5	University of Hong Kong	4	19
6	Education and Information Technologies	4	18	6	Monterrey Institute of Technology and Higher Education	4	12
7	Frontiers in Education	5	15	7	University of South Africa	4	12
8	Sustainability	4	13	8	All India Institute of Medical Sciences	4	10
9	Journal of Chemical Education	4	9	9	Raiganj Government Medical College & Hospital	3	10
10	JMIR Medical Education	5	8	10	University of Jordan Hospital	3	6

Table 4
Number and percentage of the literature within the ISCED framework.

Broad group	Fields of education	Number	Percentage
Education	Teacher training and education science	120	38.5%
Humanities and arts	Humanities	16	5.1%
Social business and law	Social and behavioral science Journalism and information Business and administration Law	26	8.3%
Science	Computing Life sciences Physical sciences	56	17.9%
Engineering manufacturing and construction	Engineering and engineering trades Manufacturing and processing	10	3.2%
Health and welfare	Medical services Medicine Dental services Nursing	62	19.9%
Services	Personal services Environmental protection	22	7.1%

experiments that investigate various determinants influencing its integration into educational practice, thereby it could help address RQ7.

The keywords constituting the fourth theme reflect the academia's considerations upon the inevitable challenges, risks and limitations posed by the rapid evolution of AI technology and its growing integration with educational practice.

4.5. Applications and effectiveness of ChatGPT in assisting higher education (RQ5)

ChatGPT has demonstrated remarkable effectiveness in a wide range of tasks. Its effectiveness stems from its capabilities to understand and respond to complex queries, provide relevant information, and generate creative contents. Applications and effectiveness of ChatGPT for assisting college teachers, students and researchers in real educational practice are summarized based on the evidence from the included 67 literature sources, as shown in Fig. 3.

4.5.1. ChatGPT as a teaching assistant

4.5.1.1. Material generation. Based-on its powerful natural language processing nature, ChatGPT have been used to assist teachers in generating useful and personalized contents with naturalness and diversification, including lesson plans, concept explanations, teaching documentations, students exercises and brainstorming questions and answers and even laboratory reports etc., thus incredibly saving time and improving teachers' working efficiency (Meron & Araci, 2023; Podlasov & Matviichuk, 2023; West et al., 2023).

By combining ChatGPT with existing question banks and specific models, it compiles personalized questions that match the corresponding abilities for different levels of student groups (Niu & Xue, 2023). A survey made by Korean researchers to evaluate teachers' perceptions of readings from two distinct sources revealed that ChatGPT-developed readings were perceived as similar in terms of the target passages' flow and expressions, indicating ChatGPT's effectiveness in L2 teaching (Shin & Lee, 2023). In mathematics and STEM courses, ChatGPT-generated final-exam questions could rival human-written questions in terms of quality, appropriateness and difficulty, highlighting its potential to streamline the production and assessment of large-scale exams and to significantly reduce faculty's workload from days to mere seconds (Drori et al., 2023, pp. 3947–3955).

4.5.1.2. Assessment and evaluation. ChatGPT has shown significant value in assisting teachers with homework assessments through scoring, giving feedback, providing detailed instructions, etc. This will incredibly relieve teachers' workload given fact that delivering timely and personalized feedback is particularly time-consuming in large classes in higher education (Azaiz et al., 2023).

A mixed-methods investigation conducted in a research university in southern China suggested that ChatGPT facilitated teachers' assessments in academic writing by cultivating better understanding of teachers' evaluation, encouraging students to make judgement and think independently regarding revisions. ChatGPT's efficiency in scoring students' writing was confirmed by the intraclass correlation coefficient with moderate to good consistency with teacher evaluations (Lu et al., 2024). In a study exploring ChatGPT's capability in analyzing students'

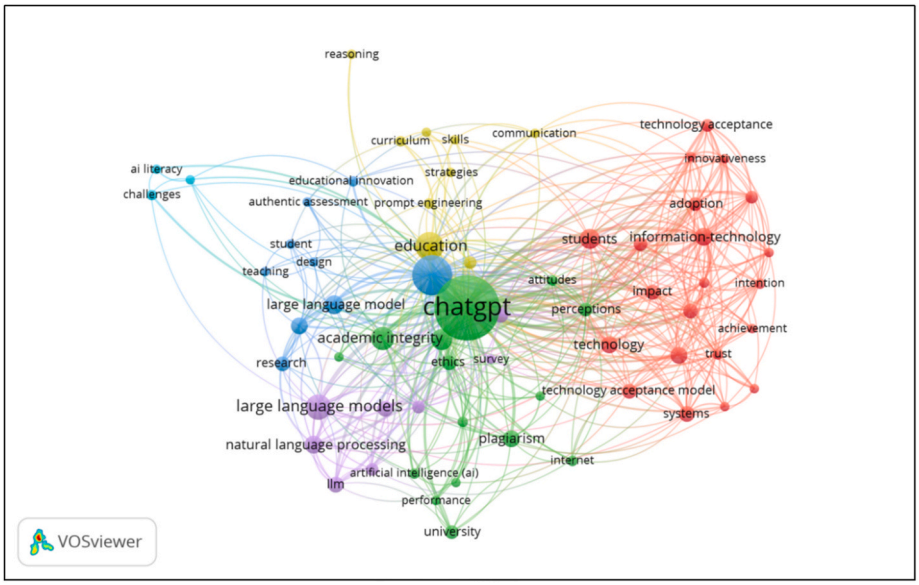


Fig. 2. The network of co-occurring keywords.

Table 5
Major themes based on keywords clusters.

Colored areas	keywords within the areas	Major themes
yellow and dark blue areas	higher education, education, student, teaching, learning, design, assessment, research, large language model, educational innovation, curriculum, skills, strategies, prompt engineering, communication, reasoning	Applications of ChatGPT in teaching, learning and research in higher education
purple area	natural language processing, large language models, LLM, medical education, survey, machine learning	Testing the performance of ChatGPT across different disciplines
red area	students, information technology, adoption, technology acceptance, user acceptance, achievement, trust, system, unified theory, innovativeness, perceptions, perceived ease, variables, internet, determinants, model, impact	Acceptance of ChatGPT in higher education
green and light blue areas	challenges, risks, AI literacy, authentic assessment, educational innovation, plagiarism, academic integrity, ethics, management, internet, writing, performance, university, chatbots	Impact of ChatGPT on higher education

course evaluation comments, there was a high level of agreement between human and machine identified themes with the range from 0.71 to 0.82 for course-related topics, and ChatGPT users demonstrated higher agreement with one another than with the evaluators. On average, teachers took more than twice the time (27.50 ± 15.00 min) to analyze their data than ChatGPT users did (10.50 ± 1.00 min), greatly surpassing human instructors in evaluation efficiency while maintaining accuracy and consistency (A. Fuller, Morbitzer, Zeeman, Persky A. CSavage, & McLaughlin, 2024).

4.5.1.3. Teaching approach enhancement. ChatGPT was adopted by educators to enhance and diversify teaching approaches. Studies have highlighted ChatGPT's capabilities in enhancing teaching efficiency, creating an engaging learning environment and personalizing learning process in various online learning modes. In the field of remote learning, ChatGPT holds promising potentials (Sisilitsyn & V. Osadchyi, 2023). It is revealed that generative AI tools are crucial in developing massive MOOC virtual classrooms where teacher can design and deliver personalized and enriching educational experiences (Ruiz-Rojas, Acosta-Vargas, De-Moreta-Llovet, & Gonzalez-Rodriguez, 2023).

In an ESL classroom at a Malaysian public university, ChatGPT acts as a virtual teacher, offering assistance by providing outlines for speech drafts, editing and revising students' work, and engaging in conversations on a variety of topics. Based on both qualitative and quantitative data, the findings revealed that ChatGPT was significantly effective in creating a learning environment to enhance students' confidence and speaking skills in flipped classroom settings (Muniandy & Selvanathan, 2024).

4.5.1.4. Teacher education and training. ChatGPT contributes to more equitable, open and accessible teacher education opportunities. It can provide materials and support mechanisms, previously available only to the privileged and economically well-off, to all teachers and student teachers. Different models can be integrated in AI-enhanced teacher training programs to provide customized teaching strategies and tools, enhancing teachers' teaching without compromising fundamental educational values (van den Berg & du Plessis, 2023). While using ChatGPT, teachers are supposed to be cautious in evaluating and critiquing both the existing and generated materials, developing critical thinking and problem-solving skills.

Based on a survey on 216 college teachers, Lan (2024) conceptualized three AI-enhanced teacher training models in which ChatGPT promotes teachers' career development as a navigator, collaborator and inventor. Teacher training programs should combine the three models to provide diverse teaching strategies and instruments by organizing discussion groups to solve practical cases, simulating teaching environments, providing instant feedback and encouraging creative project design with the help of new technologies.

4.5.2. ChatGPT as a learning assistant

4.5.2.1. Personalized learning experience. ChatGPT can be leveraged to provide personalized learning experiences by adapting its responses to individual students' needs, interests, and learning styles. This can help tailor the educational content to better suit each student's unique learning process. In language studies, ChatGPT can personalize language learning and improve students' productivity by providing learning

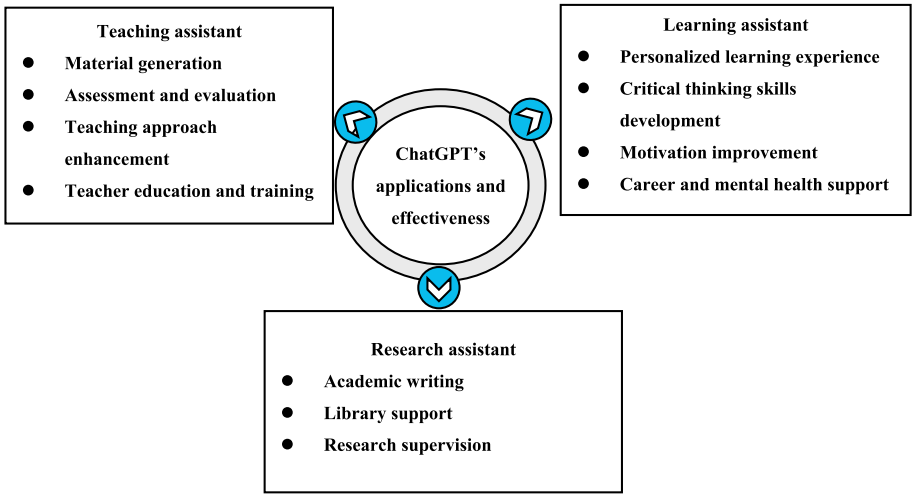


Fig. 3. ChatGPT's applications and effectiveness for teachers, students and researchers.

materials, improving writing skills and serving as a conversational partner (Nugroho, Putro, & Syamsi, 2023). ChatGPT can potentially benefit international students by providing personalized and adaptive learning opportunities. For example, language translation services can be produced to help to mitigate the cultural and language barriers faced by international students (T. Wang et al., 2023). In music education, ChatGPT was utilized to customize learning material to correspond with the distinct learning objectives and musical preference of individual students. The results of the educational experiment revealed that the experimental group exhibited a higher level of musical knowledge acquisition in comparison to the control group, indicating ChatGPT's noteworthy effectiveness for enhancing student's music learning abilities (Zhou & Kim, 2024).

4.5.2.2. Critical thinking skills development. By engaging in discussions with ChatGPT, students can develop their critical thinking and problem-solving skills. The model's ability to generate multiple perspectives and potential solutions can encourage students to think more creatively and critically (Suriano, Plebe, Acciai, & Fabio, 2025). In reverse, critically embracing ChatGPT-generated solutions helped students develop meta-cognition and critical thinking ability in reverse (Exintaris, Karunaratne, & Yuriev, 2023).

In an introductory chemistry class, practices were implemented to specifically target the enhancement of critical thinking skills. The findings revealed notable improvement in students' awareness of their critical thinking competence, with an increase from an initial 46% to a postactivity 67% students associating their use of ChatGPT with an amplified sense of self-efficacy in the application of critical thinking skills (Guo & Lee, 2023). In a study that evaluated the impact of ChatGPT on critical thinking skills among 392 medical students in Saudi Arabia, the majority of participants (92.6%) believed that incorporating ChatGPT would enhance critical thinking, with 174 strongly recognizing its positive influence. The research concluded that integrating ChatGPT into medical education had potential to foster critical thinking abilities (Almazrou et al., 2024).

4.5.2.3. Motivation enhancement. ChatGPT holds potential in promoting students' learning motivation to learn. More positive characteristics of motivation are detected in ChatGPT-based education than negative ones due to the conversational and social nature of chatbots (Hmoud et al., 2024). In EAL learning, a Multidimensional Approach Culture, Humor, and Empathy Bot (MACHE-Bot) leveraged AI to generate culturally relevant humor and empathy, thereby enhancing learner's learning outcomes and overall well-being. Five essential factors collectively contributed to the positive learning experiences: intercultural dimensions, user trust, anthropomorphism, reciprocal self-disclosure and higher engagement and motivation (Zhai, Wibowo, & Li, 2024). In AI-powered math class, students reported positive learning experiences with ChatGPT, with mean scores for perceived usefulness in comprehending mathematical concepts, problem-solving confidence, and explaining thought processes all exceeding 3.5 on a 5-point scale, which contributes to the enhancement of students' self-efficacy belief (Canonigo, 2024).

4.5.2.4. Career and mental health support. ChatGPT can provide personalized career guidance and job selection by helping students identify their strengths, interests, aptitudes, abilities and goals (Alqahtani et al., 2023). The career decision-making can be facilitated with the tailored advice on potential career paths. ChatGPT and other AI-powered tools also offers solutions to mental health problems among college students with their NLP capability (Alqahtani et al., 2023). This technology create a safe and non-judgmental environment for students to confide their emotions, and assist in the early identification of mental health. For instance, a research found that ChatGPT was considered a potential and reliable tool for educators and students interested in the topic of sexual consent. It consistently offered detailed insights on sexual consent when responding to various questions. Meanwhile it effectively provided examples of consent, including both verbal and nonverbal cues and the influence of substances (Marcantonio et al., 2023).

4.5.3. ChatGPT as a research assistant

4.5.3.1. Academic writing support. The role of AI as a research assistant was quite acceptable among researchers and editors. Its powerful text-generating capability has been notably displayed in facilitating authors in academic writing through text generation, data analysis, results interpretation, literature review, formatting and editing, and peer review (Alqahtani et al., 2023; Crawford, Cowling, & Allen, 2023; De Paoli, 2023), revolutionizing the way education and research occur. The integration of AI assistance in the academia area would ease the burden of both academics and students during the writing process and help improving the quality of academic papers (Kamal & Khaidzir, 2024).

A recent survey delved into Chinese medical students' experience in using ChatGPT to write English academic writing. Feedback from the questionnaire revealed that 92% students reported an improvement in the quality of their writing, 84% noted progress in their language skills, and 76% recognized the contribution of ChatGPT in supporting academic research. These positive results highlighted the effectiveness of large language models in augmenting the English academic writing proficiency of non-native speakers in medical education (J. Li et al.,

2024).

4.5.3.2. Library service. ChatGPT is believed to hold potential to assist researchers and facilitate librarian's work by providing quick and efficient libraries' services (Nehra & Bansode, 2024). An AI-powered chatbot integrated into the library's system can be used to provide recommendations for books or other references, which enables librarians to focus on more complex tasks and allow them to provide better service (Houston & Corrado, 2023). A real case is that a chatbot named Aisha in Zayed University Library of United Arab Emirates was designed to provide such services to students and faculty outside the regular working hours, leading to a more accessible and convenient user experience. Adding additional features like speech-to-text or text-speech transformation, it can possibly enhance the accessibility and visibility for individuals with impairments (Lappalainen & Narayanan, 2023).

4.5.3.3. Research supervision. ChatGPT's capabilities can be leveraged to enhance undergraduate and postgraduate research supervision. On the one hand, ChatGPT could support research by providing constructive feedback for students prior to submitting draft to supervisory teams, which may promote student psychological need fulfilment, autonomy, competence and relatedness(Cowling, Crawford, Allen, & Wehmeyer, 2023). On the other hand, ChatGPT has an impact on the evolution of postgraduate supervisory relationship. AI tools used in research activities play a role in shaping an enhanced supervision model, in which the teachers provide more high-level guidance and personalized feedback, leaving the postgraduate students more autonomy and independence in self-directed researches (Dai, Lai, Lim, & Liu, 2023).

4.6. Performance of ChatGPT across different disciplines(RQ6)

Empirical studies revealed that ChatGPT's performance varied significantly across disciplines, which can be reflected by the numbers of publications across difference subjects as shown in Table 4.

In the field of humanities and social sciences, the performance of ChatGPT was found to be affected by the complexity of the subject itself and the question types. When required to take college entrance exams, ChatGPT achieved excellent accuracy rates in subjects like history, economy and psychology, but performed poorly in linguistics, with linguistic complexity being a major contributing factor (Beltozar-Clemente, Diaz-Vega, Tejeda-Navarrete, & Zapata-Paulini, 2024). It was also found that in solving economics-related questions, AI tools achieved better results in numerical questions. However, their capabilities notably declines at the higher levels of the Bloom's taxonomy, such as "analyze", "evaluate" and "create" (Thanh et al., 2023).

In the field of medicine, one of the major ChatGPT-assisted areas, we synthesized the findings of nine empirical studies in which ChatGPT competed with medical students in sitting for standardized tests featuring multiple choice questions. With regard to subjects like periodontology, Otorhinolaryngology, obstetrics and gynecology, ChatGPT outperformed human candidates with above-average scores, but misinformation was occasionally observable (Danesh, Pazouki, Danesh, Danesh, & Vardar-Sengul, 2024; Li et al., 2023; Long et al., 2024). For the rest of the studies, ChatGPT failed to reach the average score of students or didn't even pass the overall pass mark (Cuthbert & Simpson, 2023; Saad, Iyengar, Kurisunkal, & Botchu, 2023; Sallam & Al-Salahat, 2023; Sarangi et al., 2023; Surapaneni et al., 2023; X. Wang et al., 2023). Therefore, ChatGPT can not replace professional doctors for the time being. When using it as a tool in clinical practice, students and clinicians should exert caution and be aware of its fallibility (Cuthbert & Simpson, 2023).

In the field of computing, we extracted five studies in which ChatGPT was used for programming courses. ChatGPT performed well in assessing student programming homework (Popovici, 2023), but it was unsatisfactory in sitting programming exams, merely comparable to a mid-range student (Ellis, Casey, & Hill, 2024; Popovici, 2023). It was unable to pass the assessments (Savelka, Agarwal, An, Bogart, & Sakr, 2023; Savelka, Agarwal, Bogart, Song, & Sakr, 2023) and unable to give the best solution in terms of object-oriented design (Cipriano & Alves, 2023). These findings indicates that ChatGPT could only assist teachers and students in the introductory or intermediate stage of computing courses due to its current insufficiency.

In conclusion, ChatGPT performed better in questions requiring low-level abilities than those requiring high-level thinking skills, regardless of humanities or scientific questions. Although it has demonstrated better performance than ChatGPT 3.5 in many studies (Danesh et al., 2024; Saad et al., 2023; Thanh et al., 2023), ChatGPT-4.0 still has significant potential for improvement in producing more accurate and pertinent content through deep learning techniques.

4.7. Adoption and acceptance of ChatGPT in higher education(RQ7)

From the three grouped clusters identified by CitNetExplorer, we selected the largest one and drilled down the longest paths, which led to Davis and Davis (1989) and Venkatesh, Morris, Davis, and Davis (2003)—creators of the Technology Acceptance Model(TAM) and the Unified Theories of Acceptance and Use of Technology (UTAUT) respectively—as the highest nodes. Major findings will be discussed according to the citation networks.

4.7.1. Evolution of TAM and its development

The original TAM model was grounded in the field of psychology with only three basic factors, namely perceived usefulness, perceived ease of use and attitude (Davis, F. D.,1986). Among the three determinants, attitude is considered dominant on the user's actual use and is influence by the other two constructs. This theory manifested itself in a research aiming to predict doctoral students' behavioral intention to use ChatGPT for writing (Zou & Huang, 2023) (see Fig. 4).

By incorporating external predictors, factors from other theories, contextual factors and usage measures, the extended TAM models were enabled to make more valid predictions to better understand the determinants of the technology acceptance (Marangunić & Granić, 2015). Rahman, Sabbir, Zhang, Moral, and Hossain (2023) added two less-explored constructs, perceived enjoyment and perceived informativeness, and believed that trust mediated the relationship between perceived enjoyment and attitude. Duong, Vu, and Ngo (2023) found that effort expectancy and knowledge sharing had a direct impact on student actual use of ChatGPT.

Abdaljaleel et al. (2024) further developed the model into a "TAME-ChatGPT" instrument specifically designed to evaluate the adoption of ChatGPT. Their results highlighted the collective influence of these factors on the successful integration of ChatGPT in higher education, including perceived ease of use, perceived usefulness, positive

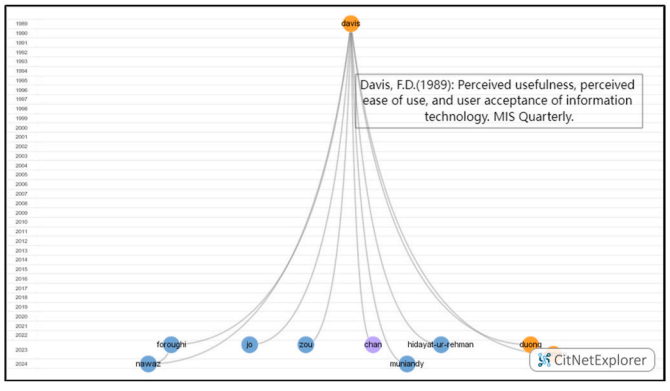


Fig. 4. Multiple longest path identified in Cluster 1 starting from Davis and Davis (1989).

attitude towards technology, social influence, behavioral/cognitive elements, low anxiety, and minimal perceived risks.

4.7.2. Evolution of UTAUT and it extensions

The earliest UTAUT framework focused on six major variables and their interplay, namely performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), behavioral intention (BI), and usage behaviour (UB) (Venkatesh et al., 2003). Duong, Bui, Pham, Vu, and Nguyen (2023) explored the impact of EE and PE on college students' intentions and actual uses of ChatGPT and concluded that their impact was positive but it will be lowered when there is an inconsistency between EE and PE (see Fig. 5).

With the development of modern technology and the increasing demand for a better examining instrument, the original UTAUT was extended into UTAUT 2 with the incorporation of three new constructs as hedonic motivation (HM), price value (PV), and habit (HA) and three moderators as age, gender and experience (Tamilmani, Rana, Wamba, & Dwivedi, 2021). It turned out that findings were consistent across studies framed with UTAUT2, revealing that performance expectancy, effort expectancy, social influence, hedonic motivation, price value, habit all exert significant influence on students' behavioral intention to use ChatGPT, while facilitating conditions, habit, and behavioral intention act as primary drivers of usage behaviour (Foroughi et al., 2023; Nawaz et al., 2024; Romero-Rodriguez, Ramirez-Montoya, Buenestado-Fernandez, & Lara-Lara, 2023; Salifu, Arthur, Arkorful, Nortey, & Osei-Yaw, 2024; Strzelecki, 2023; Strzelecki & ElArabawy, 2024).

Meta-UTAUT is another modified version of UTAUT which holds that attitude was the central mediator to behavioral intentions and usage behaviours (Dwivedi, Rana, Jeyaraj, Clement, & Williams, 2019). In an empirical studies examining Dutch college students' acceptance of ChatGPT, it was revealed that attitude and behavioral intention acted as positive predictors of usage behaviour, and anthropomorphism, design novelty, trust, performance expectancy, and effort expectancy were positive antecedents of attitude (Polyportis & Pahos, 2024).

Despite the continuous evolution and advancement of the UTAUT model, some researchers preferred to adapt the models according to their specific survey needs by introducing new variables. For instance, Hidayat-ur-Rehman and Ibrahim (2023) employed an integrated "Educator's Adoption of ChatGPT" model based on the theoretical frameworks of UTAUT and Status Quo Bias. Jo (2023) present a comprehensive model including 13 uncommon variables to understand user engagement with ChatGPT. These studies provided more comprehensive insights concerning the interplay between factors influencing the adoption and acceptance of ChatGPT.

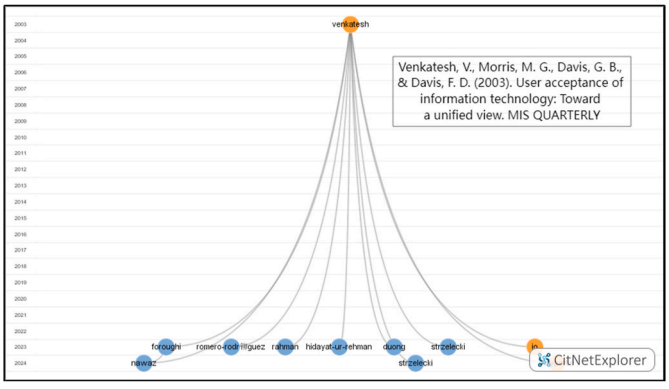


Fig. 5. Multiple longest path identified in Cluster 1 starting from Venkatesh et al. (2003).

5. Discussion

5.1. Potential benefits to higher education

The advantages of ChatGPT can be fully unleashed when it empowers higher education by acting as teaching, learning and research assistants in ways as mentioned in section 4.5, bringing numerous benefits to the field of higher education. As a conversational chatbot, based on its core technique—the generative pre-trained (GPT) model—it is capable of understanding linguistic logic, processing natural languages, and constantly adjusting and improving itself based on user inputs, thus excelling over other AI tools in producing real-time and personalized responses (Farrokhnia, Banihashem, Noroozi, & Wals, 2023).

Although researchers appeal for caution, guidance and supervisions for those who wish to incorporate this powerful tool (van den Berg & du Plessis, 2023; West et al., 2023), the incorporation of ChatGPT into higher education is believed to level the playing field for all people by opening access to teaching materials and supporting mechanisms (van den Berg & du Plessis, 2023b).

5.2. Limitations in educational practice

The remarkable capabilities of ChatGPT are sometimes overshadowed by its deficiencies in content generation, such as inaccuracy, inconsistency and hallucination. For instance, the AI generated contents required too much human prompting and editing and tended to be generic (Meron & Araci, 2023). In designing exams, ChatGPT compiled and matched questions for students of different levels with limited variety and accuracy (Niu & Xue, 2023). In assisting college Physics teacher's teaching, ChatGPT lacked comprehensive knowledge and adequate training (Podlasov & Matviichuk, 2023).

There is also a danger for learners to completely trust ChatGPT without checking the authenticity of the generated texts and thus negatively affect the acquisition of knowledge and skills (Kiryakova & Angelova, 2023). Additionally, students might possibly become overly dependent on ChatGPT for their daily tasks, which may be detrimental to the cultivation of advanced thinking skills essential for their long-term study and career development (Putra, Rangka, Aminah, & Aditama, 2023). All these limitations and risks in some ways can be destructive, posing a threat to the overall quality of higher education.

5.3. Ethical and legal risks

One of the reasons why the academia remains cautious about the application of ChatGPT in higher education derives from the concern over potential violations of academic integrity, especially those associated with academic dishonesty and plagiarism in the field of academic research and publication (Cotton, Cotton, & Shipway, 2023; Crawford, Cowling, & Allen, 2023; Jarrah, Wardat, & Fidalgo, 2023).

The legal problems resulting from use of ChatGPT is equally alarming. ChatGPT and other AI models can give rise to legal risks by producing offensive or copyrighted content, impingement of privacy, fraudulent transactions and spreading false information (Dwivedi et al., 2023) Biased or discriminated information may be produced when there is intrinsic bias in AI algorithms itself, perpetuating existing educational inequalities (Alqahtani et al., 2023).

Lastly, the release of ChatGPT has sparkled a renewed debate on the impact of disruptive technologies on human jobs and labor market (Dianova & Schultz, 2023). With the development of Generative AI technology, there is an increasingly alarming tendency that AI will gradually replace the roles of teachers or compete with college students in the already competitive job markets.

5.4. Mitigation strategies

5.4.1. Cultivation of AI literacy

Since its debut in educational stage, ChatGPT has been seen as a double-edged sword, bring astonishing potentials and unavoidable issues at the same time. In order to improve the effectiveness of ChatGPT and foster a positive AI-powered educational environment, the academic community calls for the establishment of a comprehensive framework of AI literacy which originates from the concepts of data literacy or digital literacy (Chan & Lee, 2023; Dianova & Schultz, 2023; S. Lee & Park, 2024; Sperling et al., 2024).

Lee and Park (2024) highlighted the importance of measuring users' digital literacy in fostering a healthy digital ecosystem and introduced a framework for AI literacy with five essential factors: technical proficiency, critical evaluation, communication proficiency, creative application and ethical competence. Farrelly and Baker (2023) proposed an AI literacy framework with three basic components: to acquire an understanding of the basic concept of AI, to use AI concepts for evaluation and to use AI concepts for understanding the world through problem solving. In the future, AI literacy is not only a technical expertise but rather a holistic understanding that aligns with humanistic values and ethical considerations (Tzirides et al., 2024).

For higher education institutions, they should focus on cultivating AI literacy to equip students with knowledge, skills and competencies in dealing with AI technology (Dai et al., 2023). Universities should teach students to understand the basics of how AI tools work, including their potential biases and risks (Alm & Watanabe, 2023). This can help students develop a deeper understanding of AI and its potential impact on their field of study.

In practice, curriculum should combine practical knowledge about AI's applications with hands-on experience with AI tools and techniques. Human efforts are indispensable in supporting students' learning experience in AI-assisted educational context. With human teaching assistant and ChatGPT working collaboratively, the learning environment promoted an active, hands-on instructional style, which is beneficial for teaching highly technical concepts(Pahi, Hawlader, Hicks, Zaman, & Phan, 2024).

Integrating information technology curriculum is also considered helpful for reducing technostress incurred by a lack of digital learning skills (Lin & Yu, 2024). Similarly, ethical topics such as data privacy, biased information, reliability of AI outputs, and steps to mitigate the risks from these implications should be integrated to ensure responsible use of AI (Tzirides et al., 2024), which are crucial for fostering a critical and analytical mindset in students. A trans-disciplinary pedagogy was considered specifically necessary for university students' AI literacy education. By bridging humanities with STEM subjects, digital humanities education will "promote responsible use of generative AI and encouraging critical reflection on AI's socio-cultural embeddedness"(Dianova & Schultz, 2023).

It is equally important for higher institution to provide AI literacy training and support for faculty members, enabling them to use ChatGPT effectively and address the potential challenges and limitations of AI technologies. To achieve this, education scientists need to reach a broader consensus of what the theoretical knowledge of AI literacy regarding teacher education should entail. During this exploring process teachers need to be involved in to ensure a contextualized AI literacy in authentic teaching with ethical considerations (Sperling et al., 2024). In reverse, offering AI literacy education to teachers will help students become successful users and creators of AI. Hur (2024) noted that integrating hands-on activities into AI literacy education can foster preservice teachers' confidence and positive attitudes toward using AI in education. Consequently, teachers' understanding of AI can be useful to promote creativity and problem solving with AI among students. Collaborative, problem-solving and hands-on activities are significant for both teachers' and students' AI education.

5.4.2. Responsibilities of stakeholders

To address the shortcomings of ChatGPT and the ethical and risk considerations in its application within higher education, it is imperative that collaborative efforts involving various stakeholders to be undertaken to ensure the quality and equity of higher education. This collective approach will guide the development and implementation of AI technologies in a manner that is both pedagogically sound and ethically responsible.

AI inventors should prioritize the upgrading of AI products to ensure that they can effectively fulfill the role of a facilitator in actual educational scenarios. They should collaborate with educators to leverage AI technologies with specific goals to improve learning and teaching on a large scale (Zhang & Aslan, 2021). This involves not only refining the technology to better align with pedagogical goals but also addressing the ethical considerations and privacy concerns that arise with the integration of AI in education.

In academia, in view of the ethical concerns, editorial policies have been established by publishers to regulate the use of ChatGPT in academic writing, with a central principle being ChatGPT is not entitled to any real authorship (Crawford, Cowling, Ashton-Hay, et al., 2023; Lodge, Thompson, & Corrin, 2023; Rahimi & Abadi, 2023). The academia is supposed to delve deeper into this subject, striving to come up with additional solutions to tackle critical challenges associated with AI-assisted academic research. This includes addressing ethical concerns such as privacy issues, algorithmic bias, and academic dishonesty, thereby contributing to the responsible advancement of educational technology.

Policymakers need to make policies and take measures to ensure a legal and health environment for tech-assisted educational environment. For example, concrete measures could include developing and enforcing regulations that govern the use of AI-powered technology in education, providing fund and support for the development of AI-powered educational tools. Administrative stakeholders can facilitate workshops, seminars or informational sessions that highlight the benefits of using AI for educational purposes (Jo, 2023). In addition, governments can work with educational institutions to develop policies and guidelines for the integration of technology into the curriculum.

It is also imperative for all stakeholders to demystify ChatGPT to enhance people's willingness to use this new technology. Since trust is a key factor for the users' acceptance and adoption of AI (Polyportis & Pahos, 2024; Rahman et al., 2023), it's vital for the marketers to deliver an approachable and non-intimidating user experience, thereby not overwhelming the users (Jo, 2023). To build an environment beneficial to user's engagement, educators, AI designers and developers should aim to emphasize ChatGPT's role as a supportive capabilities and foster more positive user attitudes towards AI.

In order to address all the challenges and limitations presented by ChatGPT, experts from different fields should collaboratively work on developing guidelines and best practices for the responsible use of ChatGPT, ensuring that it aligns with educational goals and students' well-being. In such interdisciplinary efforts, both teachers and students can harness the full potential of ChatGPT to revolutionize higher education while mitigating potential threats and risks.

6. Conclusion

6.1. Major findings

Since the emergence of ChatGPT in late 2022, the number of publications exploring ChatGPT-assisted higher education has surged up owing to its revolutionary impact on all aspects of higher education. It serves as a transformative tool for traditional teaching, learning and research practices. By leveraging its human-like text generation capabilities, it has fostered a more interactive and efficient educational environment across different subjects. Models like TAM and UTAUT held their validity in predicting and examining the variables influencing

users' intention and usage of ChatGPT. Meanwhile, concerns about its limitations and potential threats to higher education have never stopped, urging people to take on measures to mitigate its risks while maximize its strengths.

6.2. Limitations

Despite the significant achievements of this literature review, we acknowledges its limitations. One is that we confined our research subject to ChatGPT, the most representative LLM. The inclusion of competitors like Google's Bard or other Gen AI products in this review could alter the findings in several ways. Firstly, it would broaden the scope of the analysis to include a wider range of AI-powered tools and their capabilities. This could lead to the identification of additional strengths and weaknesses in ChatGPT compared to its competitors, revealing new opportunities and challenges for the use of AI in education and research. Secondly, the inclusion of competitors could also affect the overall evaluation of ChatGPT's performance, as the findings may now be relative to the performance of other similar products.

Another one is that this literature review has not conducted enough quantitative analysis to reveal deeper rationales of the superficial phenomena. It would be more convincing to compare the performance of ChatGPT in different subjects if more experimental data or test results were provided. It is necessary for future research to consider a meta-analysis review by adopting more rigorous methods to conduct more in-depth analysis. Researchers could also analyze usage data to examine the impact of these tools on student learning outcome. This would provide a more comprehensive understanding of the effectiveness of AI-powered tools in education and research.

6.3. Implications for future applications and research

In the future, applications and research should focus on exploring interdisciplinary collaboration to address the multifaceted challenges posed by ChatGPT as mentioned in this article. This includes addressing its inherent deficiencies, mitigating the ethical implications, and harnessing potential of ChatGPT to revolutionize teaching and learning practice.

Investigations into the perception and acceptance of AI technologies must continue, with a particular emphasis on understanding the dynamics of different determinants and their impacts on the usage of AI (Abdaljaleel et al., 2024; Amarathunga, 2024; Liu & Zhang, 2024). Research in this realm should carry out more empirical experiments that based on effective research methods, in an effort to examine student learning outcomes and offer insights into measures to enhance the integration of AI into higher education. Meanwhile, the adoption of ChatGPT should be analyzed in different disciplinary contexts to elucidate the nuances of how ChatGPT performance across various disciplines, which will help tailor educational applications to meet the specific needs and challenges of different academic fields.

As AI technologies advance, it is imperative that future applications and research delves into the follow-up evolutionary versions of ChatGPT and multimodal generative AI tools, such as AI-powered chatbot with complex human-computer interface, text-to-image and text-to-video AI tools and pedagogical agents (Amarathunga, 2024; Baker, Mills, Mcdonald, & Wang, 2023). By combining text, images, audio and video together, these AI tools can create more immersive, engaging and personalized learning environments for teachers and students.

The legal and ethical considerations concerning AI use in higher education must also be a priority (Amarathunga, 2024). Collaborative efforts among AI developers, educational institutions, and policymakers are essential to address these complex issues and shape the future of AI in higher education responsibly and effectively. Future research should provide insights into more practical methods in dealing with these issues.

Statements on open data and ethics

Since this study does include any participant, informed consent was not necessary. The data can be obtained by sending request e-mails to the corresponding author.

Availability of data and material

We make sure that all data and materials support our published claims and comply with field standards.

Ethics approval statement

The study was approved by the institutional review board of Shandong Foreign Trade Vocational College. All researchers can provide written informed consents.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Kimi, a large language model, to check text for spelling, grammar errors and clarity. After using this tool, the author reviewed and edited the content as needed and took full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Abbreviations

Gen AI	Generative Artificial Intelligence
LLM	large language model
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
DOI	Digital Object Identifier
AERA	American Educational Research Association
ISCED	International Standard Classification of Education
UTAUT	Unified Theory of Acceptance and Use of Technology
TAM	Technology Acceptance Model
PE	performance expectancy
EE	effort expectancy
SI	social influence
FC	facilitating conditions
BI	behavioral intention
UB	usage behaviour

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2024.100336.

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*Ultimately, 67 highly-rated articles (marked with asterisks in the references) were selected for further evidence extraction and analysis to answer RQ5 to RQ7.

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